

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0276

Instituto Tecnológico De Ponce 138 kV
January 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0276. The Interconnection Customer proposes a 81.2 MW wind facility interconnecting at the Instituto Tecnológico De Ponce 138 kV substation in the Virginia area. The request was received on October 11, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 191 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,562,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Riva Road to Shingletown 345 KV, the Taber to Dixleta 345 KV L loads to 118.7% of its 1,108 MVA rating. Project DFAX: 18.0%.

For the loss of Taber to Dixleta 345 KV, the High Ridge to N Pole 138 KV loads to 117.3% of its 372 MVA rating. Project DFAX: 8.4%.

The Wilkens to Highgrove 138 KV is overloaded to 116.7% of its 307 MVA emergency rating for the loss of the Dixleta to Taber 345 KV. The project contributes 5.8% DFAX (4.7 MW) to this overload.

Beltsville to Bunker Hill 138 KV: 112.8% loading (431 MVA) following the outage of the Taber to Glenannan 345 KV; DFAX = 5.6%.

For the loss of Taber Substation to Glenannan 345 KV, the Lak Creek to Sierra Pacific Lumber 138 KV loads to 112.2% of its 445 MVA rating. Project DFAX: 5.3%.

For the loss of Sierra Pacific Lumber 138 kV/1 kV TX, the Taber to Glenannan 345 KV loads to 111.4% of its 1,198 MVA rating. Project DFAX: 20.0%.

Jerome to High Ridge 138 KV: 110.6% loading (429 MVA) following the outage of the Taber to Dixleta 345 KV; DFAX = 8.4%.

For the loss of Taber to Glenannan 345 KV, the Bunker Hill to Nasa 138 KV loads to 110.5% of its 474 MVA rating. Project DFAX: 5.6%.

The Wheelabrator Baltimore Refuse to Westport 138 KV L is overloaded to 107.8% of its 278 MVA emergency rating for the loss of the Taber to Dixleta 345 KV L. The project contributes 5.9% DFAX (4.8 MW) to this overload.

Sierra Pacific Lumber 138/1kV XFMR: 106.5% loading (747 MVA) following the outage of the Glenannan to Taber Substation 345 KV; DFAX = 7.0%.

Sierra Pacific Lumber 345/1kV TX: 106.5% loading (747 MVA) following the outage of the Taber to Glenannan 345 KV; DFAX = 7.0%.

Wilkens to Wheelabrator Baltimore Refuse 138 KV Line: 106.4% loading (373 MVA) following the outage of the Taber to Dixleta 345 KV; DFAX = 5.8%.

For the loss of Taber to Glenannan 345 KV L, the Adrian Energy Associates Llc 345/138kV Transformer loads to 103.9% of its 519 MVA rating. Project DFAX: 5.3%.

Sunbright to Fairland 138 KV: 103.6% loading (513 MVA) following the outage of the Glenannan to Taber 345 KV L; DFAX = 7.5%.

Contr Costa to Libby 138 KV: 103.2% loading (296 MVA) following the outage of the N Anna 345 kV/138 kV Autotransformer; DFAX = 5.8%.

For the loss of Glenannan to Taber 345 KV, the Sunbright to Colesville Substation 138 KV loads to 101.8% of its 577 MVA rating. Project DFAX: 7.5%.

For the loss of Dixleta to Taber 345 KV, the Taber 345/138kV Transformer Bank loads to 100.5% of its 767 MVA rating. Project DFAX: 5.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Taber to Dixleta 345 KV L	Rebuild line with higher-capacity conductor	\$31,425,000	1% (\$228,000)
North Pole to High Ridge 138 KV	Rebuild line with higher-capacity conductor	\$1,948,000	1% (\$16,000)
Wilkens Substation to Highgrove 138 KV Line	Rebuild line with higher-capacity conductor	\$1,003,000	1% (\$10,000)
Beltsville to Bunker Hill 138 KV Line	Rebuild line with higher-capacity conductor	\$965,000	1% (\$9,000)
Lak Creek to Sierra Pacific Lumber 138 KV	Rebuild line with higher-capacity conductor	\$2,927,000	1% (\$37,000)
Taber to Glenannan 345 KV	Rebuild line with higher-capacity conductor	\$24,680,000	1% (\$183,000)
Jerome to High Ridge 138 KV	Rebuild line with higher-capacity conductor	\$1,853,000	1% (\$15,000)
Bunker Hill to Nasa 138 KV	Rebuild line with higher-capacity conductor	\$2,244,000	1% (\$20,000)
Wheelabrator Baltimore Refuse to Westport 138 KV	Reconductor line	\$153,000	1% (\$1,000)
Sierra Pacific Lmber 138-1 kV TX	Replace transformer with higher-rated unit	\$6,015,000	1% (\$52,000)
Sierra Pacific Lumber 345/1kV TX	Replace transformer with higher-rated unit	\$17,623,000	1% (\$151,000)
Wilkens to Wheelabrator Baltimore Refuse 138 KV Line	Reconductor line	\$615,000	1% (\$6,000)
Adrian Energy Associates Llc 345/138kV Transformer	Replace transformer with higher-rated unit	\$15,879,000	4% (\$709,000)
Sunbright to Fairland 138 KV	Reconductor line	\$181,000	1% (\$1,000)
Libby to Contra Costa 138 KV	Reconductor line	\$1,997,000	100% (\$1,997,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sunbright to Colesville 138 KV	Reconductor line	\$435,000	1% (\$3,000)
Taber 345/138kV Transformer Bank	Replace transformer with higher-rated unit	\$13,686,000	1% (\$124,000)
	Total allocated to project		\$3,562,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.